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标题	Naphthalene diimides with improved solubility for visible light photoredox catalysis
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摘要（原文节选）

Five core-substituted naphthalene diimides bearing two dialkylamino groups were synthesized as potential visible light photoredox catalysts and characterized by methods of optical spectroscopy and electrochemistry in comparison with one unsubstituted naphthalene diimide as reference. The core-substituted naphthalene diimides differ by the alkyl groups at the imide nitrogens and at the nitrogens of the two substituents at the core in order to enhance their solubility in DMF and thereby enhance their photoredox catal ⋯

数据表 · Table 2（14 行，含数值行 12 行）

Photoredox catalytic conversions of diethyl 2-bromomalonate (13) with 1-octanal (12) and yields of product 14 by different photoredox catalysts (X).

Entry	X	mol %	LED	solvent	Conversion (%) of 13	Yield (%) of 14
1	1	5	387	DMF/CH2Cl2 1:1	65	25
2a	1	5	387	DMF/CH2Cl2 1:1	87	45
3	1	2.5	387	DMF/CH2Cl2 1:1	67	32
4	1	1.25	387	DMF/CH2Cl2 1:1	67	30
5a	1	1.25	387	DMF/CH2Cl2 1:1	99	60
6	–	–	387	DMF/CH2Cl2 1:1	24	18
7	1	5	387	CH2Cl2	30	n.d.
8	1	5	387	DMF	69	44
9	–	–	387	DMF	80	58
10	6	0.1	520	DMF/CH2Cl2 1:1	77	69
11	6	0.1	597	DMF/CH2Cl2 1:1	46	43
12	6	0.1	637	DMF/CH2Cl2 1:1	n.d.	n.d.

脚注（逐字）：a40 mol % 15.

正文对应句（逐字）：Photocatalysis couples the physical process of light absorption to an organic-chemical reaction by means of time, space and energetics. In order to apply visible light for photocatalysis despite its rather low energy this coupling requires to be mediated by a sensitizing species – a photocatalyst. I

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